Type MS · Explain ultrafiltration and selective reabsorption in the nephron

Answer 1 of 13 · Question 6 part a.iii · 4 marks

2021 JUN · Paper 01 · extract 1 of 2 · find part a.iii in this mark scheme extract

number				
6 (a)	(i)	• separating different sized molecules out of the blood/only allows small molecules to pass; • under pressure;		2
	(ii)	• A has more glucose; • A has more urea; • A more amino acids;	ORA vessel B allow references to more oxygen in A and less carbon dioxide in A	3
	(iii)	• Vessel A bring blood to Bowman's capsule/glomerulus; • vessel B has narrow(er) lumen/diameter; • causes increase/higher blood pressure; • to force out molecules;		4
(b)		• glucose would pass from blood; • respiration affected/reduced; • lack of energy/ATP;		3

Total 12 marks

Type MS · Explain ultrafiltration and selective reabsorption in the nephron

Answer 1 of 13 · Question 6 part a.iii · 4 marks

2021 JUN · Paper 01 · extract 2 of 2 · find part a.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain ultrafiltration and selective reabsorption in the nephron

Answer 2 of 13 · Question 4 part a.i · 1 mark

2019 JUN · Paper 01 (regional R) · extract 1 of 3 · find part a.i in this mark scheme extract

4(a)(i)	A protein;		**(1)**

Question Number	Answer	Additional guidance	Mark
4(a)(ii)	A the molecules are too large;		**(1)**

Question Number	Answer	Additional guidance	Mark
4(a)(iii)	• more glucose in blood/filtrate; • as little/no insulin to lower blood glucose levels/glucose not converted to glycogen; • not all/too much to be reabsorbed; • some found in urine;		**(3)**

Question Number	Answer	Additionalguidance	Mark
4(a)(iv)	$\frac{25 \times 100}{53}$; 47%;	Full marks for correct final answer Allow 47.2% / 47.17 Ecf allow one mark for correct answer using incorrect figures in calculation	**(2)**

Type MS · Explain ultrafiltration and selective reabsorption in the nephron

Answer 2 of 13 · Question 4 part a.i · 1 mark

2019 JUN · Paper 01 (regional R) · extract 2 of 3 · find part a.i in this mark scheme extract

Question Number	Answer	Additional guidance	Mark
4(b)	Any five from: Advantages • transplant permanent/carried out once; • can return to normal life; • not connected to machine for long periods/no need for frequent hospital visits; Disadvantages • risk of rejection; • risk of infection (during/after surgery)/reduced immunity; • difficulties in finding donor/long waiting lists for donor;		**(5 max)**

Total 12 Marks

Type MS · Explain ultrafiltration and selective reabsorption in the nephron

Answer 2 of 13 · Question 4 part a.i · 1 mark

2019 JUN · Paper 01 (regional R) · extract 3 of 3 · find part a.i in this mark scheme extract

Paper 1: Subject Human Biology

Question Number	Answer	Additional guidance	Mark

Type MS · Explain ultrafiltration and selective reabsorption in the nephron

Answer 3 of 13 · Question 4 part a.ii · 1 mark

2019 JUN · Paper 01 (regional R) · extract 1 of 3 · find part a.ii in this mark scheme extract

4(a)(i)	A protein;		**(1)**

Question Number	Answer	Additional guidance	Mark
4(a)(ii)	A the molecules are too large;		**(1)**

Question Number	Answer	Additional guidance	Mark
4(a)(iii)	• more glucose in blood/filtrate; • as little/no insulin to lower blood glucose levels/glucose not converted to glycogen; • not all/too much to be reabsorbed; • some found in urine;		**(3)**

Question Number	Answer	Additionalguidance	Mark
4(a)(iv)	$\frac{25 \times 100}{53}$; 47%;	Full marks for correct final answer Allow 47.2% / 47.17 Ecf allow one mark for correct answer using incorrect figures in calculation	**(2)**

Type MS · Explain ultrafiltration and selective reabsorption in the nephron

Answer 3 of 13 · Question 4 part a.ii · 1 mark

2019 JUN · Paper 01 (regional R) · extract 2 of 3 · find part a.ii in this mark scheme extract

Question Number	Answer	Additional guidance	Mark
4(b)	Any five from: Advantages • transplant permanent/carried out once; • can return to normal life; • not connected to machine for long periods/no need for frequent hospital visits; Disadvantages • risk of rejection; • risk of infection (during/after surgery)/reduced immunity; • difficulties in finding donor/long waiting lists for donor;		**(5 max)**

Total 12 Marks

Type MS · Explain ultrafiltration and selective reabsorption in the nephron

Answer 3 of 13 · Question 4 part a.ii · 1 mark

2019 JUN · Paper 01 (regional R) · extract 3 of 3 · find part a.ii in this mark scheme extract

Paper 1: Subject Human Biology

Question Number	Answer	Additional guidance	Mark

Type MS · Explain ultrafiltration and selective reabsorption in the nephron

Answer 4 of 13 · Question 8 part a.i · 2 marks

2019 JUN · Paper 02 · extract 1 of 2 · find part a.i in this mark scheme extract

8 (a)(i)	• high (blood) pressure (in the glomerulus); • small molecules/named small molecule forced / filtered/pass through (into Bowmans capsule);	Do not allow protein	1 1
(ii)	Any four from: • lower concentration of glucose/oxygen in blood in renal vein; • used in respiration; • less/no urea in renal vein; • urea excreted in urine; • more carbon dioxide in renal vein; • carbon dioxide produced in respiration;	Allow reverse argument throughout	Max 4
(b) (i)	quantity of salt (in food);		1
(ii)	Any two from: • different foods used (which may affect the amount of urea produced/amount of water reabsorbed); • different people used who may be different genders/ages/ have a medical condition/different metabolism (that affects osmoregulation/water balance); • the food/drink consumed by the people prior to the investigation was not taken into account (which will influence the quantity of urea/water in urine produced);		Max 2
(iii)	use one person only (and vary salt intake)/monitor / control food/water consumption/give each person the same food (but vary salt content);	Allow valid alternatives	1

Total question 8 = 10 marks

Type MS · Explain ultrafiltration and selective reabsorption in the nephron

Answer 4 of 13 · Question 8 part a.i · 2 marks

2019 JUN · Paper 02 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain ultrafiltration and selective reabsorption in the nephron

Answer 5 of 13 · Question 6 part a.i · 2 marks

2021 JUN · Paper 01 · extract 1 of 2 · find part a.i in this mark scheme extract

number				
6 (a)	(i)	• separating different sized molecules out of the blood/only allows small molecules to pass; • under pressure;		2
	(ii)	• A has more glucose; • A has more urea; • A more amino acids;	ORA vessel B allow references to more oxygen in A and less carbon dioxide in A	3
	(iii)	• Vessel A bring blood to Bowman's capsule/glomerulus; • vessel B has narrow(er) lumen/diameter; • causes increase/higher blood pressure; • to force out molecules;		4
(b)		• glucose would pass from blood; • respiration affected/reduced; • lack of energy/ATP;		3

Total 12 marks

Type MS · Explain ultrafiltration and selective reabsorption in the nephron

Answer 5 of 13 · Question 6 part a.i · 2 marks

2021 JUN · Paper 01 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain ultrafiltration and selective reabsorption in the nephron

Answer 6 of 13 · Question 1 part a.iii · 2 marks

2022 JAN · Paper 02 · extract 1 of 2 · find part a.iii in this mark scheme extract

1	(a) (i)	• first convoluted tubule; • loop of Henlé; • second convoluted tubule;	diagram only, no labels required	3
	(ii)	44mm; 44 ÷ 650; = 0.068mm;	ecf from mp1 max 2 marks Allow 0.0676	3
	(iii)	• protein; • red/white blood cells/platelets;		2
	(iv)	glucose;		1
	(b) (i)	Any 2 from: • same person; • same activity levels; • same volume of water; • same food eaten;		2
	(ii)	710 ÷ 4; = 178 cm^3 per hour;	Allow 1 mark if division by 4 is shown Allow 177.5	2
	(iii)	• less urine production/lower urine loss on day two; • more ADH secreted on second day; • more water reabsorbed; • more water lost by sweating;	Allow reverse argument for all marking points	3

Total 16 marks

Type MS · Explain ultrafiltration and selective reabsorption in the nephron

Answer 6 of 13 · Question 1 part a.iii · 2 marks

2022 JAN · Paper 02 · extract 2 of 2 · find part a.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain ultrafiltration and selective reabsorption in the nephron

Answer 7 of 13 · Question 6 part b.i · 3 marks

2023 NOV · Paper 01 · extract 1 of 2 · find part b.i in this mark scheme extract

Question number				
6	(a)	L = Bowman's capsule; M = glomerulus; N = collecting duct; 0 = loop of Henlé ;		4
	(b) (i)	• ultrafiltration; • high blood pressure in B/glomerulus; • forces fluid/water into L/Bowman's capsule/out;		3
	(ii)	• blood cells/named; • proteins;		2
	(iii)	line labelled G pointing to proximal convoluted tubule;		1
	(c)	any five from • more sweat; • hypothalamus detects decreased water potential of blood; • ADH secreted; • increases permeability of collecting duct; • causes more reabsorption of water; • volume of filtrate reduced; • concentration of filtrate/urine increased;		5

Total 15 marks

Type MS · Explain ultrafiltration and selective reabsorption in the nephron

Answer 7 of 13 · Question 6 part b.i · 3 marks

2023 NOV · Paper 01 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain ultrafiltration and selective reabsorption in the nephron

Answer 8 of 13 · Question 6 part b.ii · 2 marks

2023 NOV · Paper 01 · extract 1 of 2 · find part b.ii in this mark scheme extract

Number				
6	(a)	L = Bowman's capsule; M = glomerulus; N = collecting duct; 0 = loop of Henlé ;		4
	(b) (i)	• ultrafiltration; • high blood pressure in B/glomerulus; • forces fluid/water into L/Bowman's capsule/out;		3
	(ii)	• blood cells/named; • proteins;		2
	(iii)	line labelled G pointing to proximal convoluted tubule;		1
	(c)	any five from • more sweat; • hypothalamus detects decreased water potential of blood; • ADH secreted; • increases permeability of collecting duct; • causes more reabsorption of water; • volume of filtrate reduced; • concentration of filtrate/urine increased;		5

Total 15 marks

Type MS · Explain ultrafiltration and selective reabsorption in the nephron

Answer 8 of 13 · Question 6 part b.ii · 2 marks

2023 NOV · Paper 01 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain ultrafiltration and selective reabsorption in the nephron

Answer 9 of 13 · Question 5 part c · 2 marks

2025 JUN · Paper 02 · extract 1 of 2 · find part c in this mark scheme extract

5 (a)	• ACE inhibitor prevents/inhibits action of enzyme; • prevents formation of angiotensin II; • less(angiotensin ii) bind with receptors; • Losartan blocks/binds to receptors; • angiotensin ii cannot bind to receptor;		4
(b)	• arteries/arterioles; • muscles in walls; • can contract/constrict; • narrow lumen; • smaller volume/space(for blood flow);		4
(c)	• in normal/healthy kidney protein not found in urine; • protein is a big molecule; • should not pass through the glomerulus/basement membrane; • damage to glomerulus/basement membrane;		2
(d)	• stronger/intense/eq; <u>contractions</u>; • (which causes) <u>muscle</u> to get thicker;	mp2 must be due to mp1	2
(e)	• detoxification occurs in the liver; • can't break down Losartan if impaired/accumulation /build-up in body/bloodstream/liver; • results in (further) liver damage/liver failure/liver diseae;		2 **Total =14**

Type MS · Explain ultrafiltration and selective reabsorption in the nephron

Answer 9 of 13 · Question 5 part c · 2 marks

2025 JUN · Paper 02 · extract 2 of 2 · find part c in this mark scheme extract

Question number	Answer	Notes	Marks
	• A=rods/cones/retina;		

Type MS · Explain ultrafiltration and selective reabsorption in the nephron

Answer 10 of 13 · Question 1 part a.iv · 1 mark

2022 JAN · Paper 02 · extract 1 of 2 · find part a.iv in this mark scheme extract

Number					
1	(a)	(i)	• first convoluted tubule; • loop of Henlé; • second convoluted tubule;	diagram only, no labels required	3
		(ii)	44mm; 44 ÷ 650; = 0.068mm;	ecf from mp1 max 2 marks Allow 0.0676	3
		(iii)	• protein; • red/white blood cells/platelets;		2
		(iv)	glucose;		1
	(b)	(i)	Any 2 from: • same person; • same activity levels; • same volume of water; • same food eaten;		2
		(ii)	710 ÷ 4; = 178 cm^3 per hour;	Allow 1 mark if division by 4 is shown Allow 177.5	2
		(iii)	• less urine production/lower urine loss on day two; • more ADH secreted on second day; • more water reabsorbed; • more water lost by sweating;	Allow reverse argument for all marking points	3

Total 16 marks

Type MS · Explain ultrafiltration and selective reabsorption in the nephron

Answer 10 of 13 · Question 1 part a.iv · 1 mark

2022 JAN · Paper 02 · extract 2 of 2 · find part a.iv in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain ultrafiltration and selective reabsorption in the nephron

Answer 11 of 13 · Question 4 part a.iii · 3 marks

2019 JUN · Paper 01 (regional R) · extract 1 of 3 · find part a.iii in this mark scheme extract

4(a)(i)	A protein;		**(1)**

Question Number	Answer	Additional guidance	Mark
4(a)(ii)	A the molecules are too large;		**(1)**

Question Number	Answer	Additional guidance	Mark
4(a)(iii)	• more glucose in blood/filtrate; • as little/no insulin to lower blood glucose levels/glucose not converted to glycogen; • not all/too much to be reabsorbed; • some found in urine;		**(3)**

Question Number	Answer	Additionalguidance	Mark
4(a)(iv)	$\frac{25 \times 100}{53}$; 47%;	Full marks for correct final answer Allow 47.2% / 47.17 Ecf allow one mark for correct answer using incorrect figures in calculation	**(2)**

Type MS · Explain ultrafiltration and selective reabsorption in the nephron

Answer 11 of 13 · Question 4 part a.iii · 3 marks

2019 JUN · Paper 01 (regional R) · extract 2 of 3 · find part a.iii in this mark scheme extract

Question Number	Answer	Additional guidance	Mark
4(b)	Any five from: Advantages • transplant permanent/carried out once; • can return to normal life; • not connected to machine for long periods/no need for frequent hospital visits; Disadvantages • risk of rejection; • risk of infection (during/after surgery)/reduced immunity; • difficulties in finding donor/long waiting lists for donor;		**(5 max)**

Total 12 Marks

Type MS · Explain ultrafiltration and selective reabsorption in the nephron

Answer 11 of 13 · Question 4 part a.iii · 3 marks

2019 JUN · Paper 01 (regional R) · extract 3 of 3 · find part a.iii in this mark scheme extract

Paper 1: Subject Human Biology

Question Number	Answer	Additional guidance	Mark

Type MS · Explain ultrafiltration and selective reabsorption in the nephron

Answer 12 of 13 · Question 3 part c.iii · 3 marks

2023 JUN · Paper 02 · extract 1 of 2 · find part c.iii in this mark scheme extract

Question number				
3	(a)	Any two from: • diseases caused by pathogens/bacteria/viruses; • diseases can be transferred from one person to another; • diabetes caused by failure to produce insulin/not caused by pathogen;		2
	(b) (i)	• 25%/1 in 4 don't know; • $\frac{25 \times 5\,000\,000;}{100}$ OR $\frac{5\,000\,000;}{4}$ • = 1.25 (million);	ecf max 1 mark for correct final answer to a calculation	3
	(ii)	• $\frac{10 \times 5\,000\,000;}{100}$ • = 0.5 (million);	ecf max 1 mark for wrong calculation but correct answer to the calculation	2
	(c) (i)	glucose;		1
	(ii)	Any two from: • reference to change in the level of fluid in the eye/ear/leakage of blood into the eye; • blood vessels/neurones in ear/cochlea/eye/retina may be damaged; • reduced blood flow to eye/ear/less nutrients/oxygen delivered;		2
	(iii)	Any three from: • glucose/sugar in filtrate/urine; • water moves from cells/tissues into filtrate/urine; • reference to water potential in filtrate/kidney being lower than in surrounding tissue/less water reabsorbed;	Allow reference to water moving from high to low water potential	3
				Total 13

Type MS · Explain ultrafiltration and selective reabsorption in the nephron

Answer 12 of 13 · Question 3 part c.iii · 3 marks

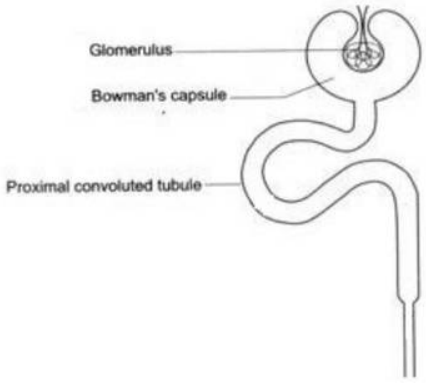
2023 JUN · Paper 02 · extract 2 of 2 · find part c.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain ultrafiltration and selective reabsorption in the nephron

Answer 13 of 13 · Question 9 part c · 4 marks

2020 NOV · Paper 01 (regional R) · extract 1 of 2 · find part c in this mark scheme extract

9 (a) (i)	• evidence of 23500 – 4000 (= 19500) (1) • 19500 ÷ 4000 (x 100)(1) • 488(%)	Allow full marks for correct final answer. Allow 487.5(%) Allow ecf if incorrect answer from mp1 – max 2 marks	3
(ii)	other lifestyle factors/genetic predisposition could have an influence/named lifestyle factor(1)	Allow gender/age	1
(b)	A diagram similar to:  	One mark for diagram. One mark for each correct label.	4
(c)	any four from • ultrafiltration does not take place(1) • in glomerulus(1) • less/no waste products/named waste product/water/glucose pass through to tubule(1) • greater volume of waste in blood/more water in blood(1) • glucose/ions not reabsorbed(1) • glucose found in urine(1) OR any four from • blood not filtered effectively/ultrafiltration ineffective(1) • in glomerulus (1)		4

Type MS · Explain ultrafiltration and selective reabsorption in the nephron

Answer 13 of 13 · Question 9 part c · 4 marks

2020 NOV · Paper 01 (regional R) · extract 2 of 2 · find part c in this mark scheme extract

	• larger molecules/named larger molecules pass through to tubule/Bowmans capsule(1) • not all glucose/ions reabsorbed(1) • glucose/protein found in urine/urine more concentrated(1)		

Total for Question 9 = 12 marks